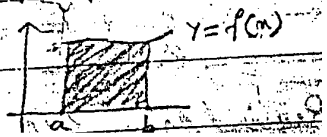


Quadrature

The process of finding the area of a plane region

- ① Area of the region bounded by the curve $y = f(x)$ the x -axis, $x = a$ and $x = b$

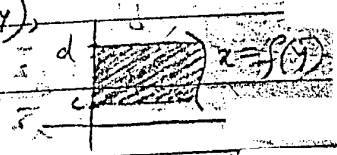
$$A = \int_a^b y \, dx$$



- ② Area of the region bounded by the curve $x = f(y)$, the y -axis, and $y = c$ and $y = d$

$$A = \int_c^d x \, dy$$

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- ③ Parametric Form $x = f(t)$ $y = g(t)$

we shall use $\int y \, dx = \int_{t_1}^{t_2} g(t) f'(t) \, dt$

- ④ Area bounded by the curve $r = f(\theta)$

and from $\theta = \alpha$ to $\theta = \beta$

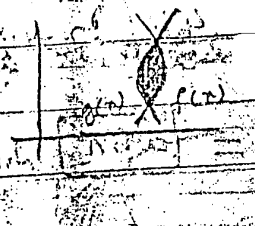
$$\int_{\alpha}^{\beta} \frac{1}{2} r^2 \, d\theta$$



- ⑤ Area between two curves

$$\int_a^b [g(x) - f(x)] \, dx$$

($g(x) > f(x)$)

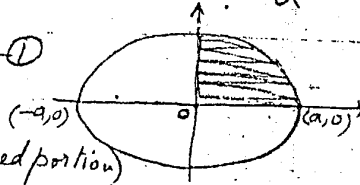


7-5-2

Ex 7.5

Q No 1. Area of the region bounded by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{--- (1)}$$



Required Area = 4 (Area of shaded portion)

$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$y^2 = \frac{b^2}{a^2} (a^2 - x^2)$$

$$y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$$

$$A = \int_a^b y \, dx = 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} \, dx$$

$$= \frac{4b}{a} \int_0^a \sqrt{a^2 - x^2} \, dx$$

$$= \frac{4b}{a} \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) \right]_0^a$$

$$= \frac{4b}{a} \left[\frac{a}{2} \sqrt{a^2 - a^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{a}{a} \right) - \left(0 + \frac{a^2}{2} \sin^{-1} (0) \right) \right]$$

$$= \frac{4b}{a} \left[0 + \frac{a^2}{2} \left(\frac{\pi}{2} \right) - 0 - 0 \right] = \frac{4b a^2 \pi}{a \cdot 4}$$

$$= \boxed{ab\pi} \text{ Ans. (Area of Ellipse)}$$

2nd Method



$$x = a \cos \theta, \quad y = b \sin \theta, \quad dx = -a \sin \theta \, d\theta$$

$$\int_0^{\pi/2} b \sin \theta (-a \sin \theta) \, d\theta$$

IIIrd Method



$$\int_a^b y \, dx$$

IVth Method



$$x = a \cos \theta, \quad y = b \sin \theta, \quad dx = -a \sin \theta \, d\theta$$

$$\int_0^{\pi/2} b \sin \theta (-a \sin \theta) \, d\theta$$

Q2. Area of the region bounded by $y = \ln x$, $x = a$, $x = b$

$$A = \int_a^b y \, dx = \int_a^b \ln x \, dx$$

$$\text{Area} = \int_a^b \frac{1}{x} \ln x \, dx.$$

Integration by parts.

$$= \left[\ln x (x) \right]_a^b - \int_a^b \frac{1}{x} (x) \, dx$$

$$= b \ln b - a \ln a - \int_a^b 1 \, dx.$$

$$= b \ln b - a \ln a - \left[x \right]_a^b$$

$$= b \ln b - a \ln a - (b - a)$$

$$A = b \ln b - a \ln a - b + a.$$

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(Q3.) Area of the region bounded by $xy = c^2$, x -axis, $x=a$, $x=b$

$$\text{Area} = \int_a^b y \, dx = \int_a^b \frac{c^2}{x} \, dx$$

$$= c^2 \int_a^b \frac{dx}{x} = c^2 \left[\ln x \right]_a^b$$

$$= c^2 (\ln b - \ln a) = c^2 \ln \left(\frac{b}{a} \right) \text{ Ans.}$$

Q4.

Area of the region bounded by parabola $y^2 = 4ax$ & latus rectum

$$y^2 = 4ax.$$

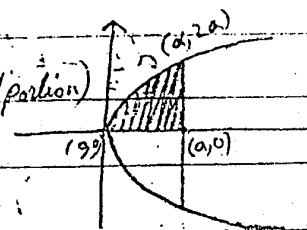
$$y = \pm 2\sqrt{ax}$$

Required Area = 2 (shaded portion)

$$\text{Area} = \int_a^b y \, dx = 2 \int_0^a 2\sqrt{ax} \, dx$$

$$= 4 \int_0^a \frac{1}{\sqrt{a}} x^{\frac{1}{2}} \, dx = 4 \frac{1}{\sqrt{a}} \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^a$$

$$= 4 \frac{1}{\sqrt{a}} \left(\frac{a^{\frac{3}{2}}}{\frac{3}{2}} - 0 \right) = \frac{8a^2}{3} \text{ Ans.}$$



7.5-4

Q.5. Area of the region bounded by $y = c \cosh\left(\frac{x}{c}\right)$, x -axis, $x=a$, $x=b$

$$\int_a^b y dx = \int_a^b c \cosh\left(\frac{x}{c}\right) dx$$

$$= c \left| \frac{\sinh \frac{x}{c}}{1/c} \right|_a^b = c^2 \left[\sinh\left(\frac{b}{c}\right) - \sinh\left(\frac{a}{c}\right) \right]$$

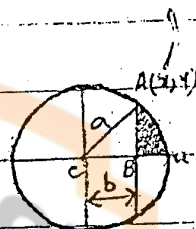
~~Ans~~
Q.6 Taking centre of disc as origin

and the chord $\parallel$ to y -axis, then

curve $x^2 + y^2 = a^2$ Required Area

$$y = +\sqrt{a^2 - x^2}$$

= 2(Shaded)



a is radius
 c is centre $(0,0)$

$$(x-h)^2 + (y-k)^2 = r^2$$

$$x^2 + y^2 = a^2$$

$$A = \int_a^b y dx = 2 \int_b^a \sqrt{a^2 - x^2} dx$$

$$= 2 \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right]$$

$$= 2 \left[\frac{a}{2} \sqrt{a^2 - a^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{a}{a} \right) - \frac{b}{2} \sqrt{a^2 - b^2} - \frac{a^2}{2} \sin^{-1} \left(\frac{b}{a} \right) \right]$$

$$= 0 + \frac{a^2}{2} \left(\frac{\pi}{2} \right) - \frac{b}{2} \sqrt{a^2 - b^2} - \frac{a^2}{2} \sin^{-1} \left(\frac{b}{a} \right)$$

$$A = \frac{a^2 \pi}{2} - b \sqrt{a^2 - b^2} - a^2 \sin^{-1} \left(\frac{b}{a} \right)$$

Ans.

Q.7

$$3ay^2 = x(x-a)$$

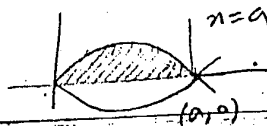
Put $y = -y$

No change $\therefore$ Curve is symmetric w.r.t x -axis

$\therefore$ boundary conditions are not given

Put $y = 0$

$$x(x-a) = 0 \Rightarrow x = 0 \quad x = a$$



7.5-5

$$3ay^2 = x(x-a)^2$$

$$\Rightarrow y^2 = \frac{x(x-a)^2}{3a}$$

$$y = \sqrt{\frac{x(x-a)^2}{3a}}$$

$$\Rightarrow y = \frac{\sqrt{x}(x-a)}{\sqrt{3a}}$$

$$\text{Area} = \int_a^b y \, dx = \int_0^a \frac{\sqrt{x}(x-a)}{\sqrt{3a}} \, dx$$

$$= \frac{2}{\sqrt{3a}} \int_0^a (x^{3/2} - ax^{1/2}) \, dx$$

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$$= \frac{2}{\sqrt{3a}} \left[\frac{x^{5/2}}{5/2} - a \frac{x^{3/2}}{3/2} \right]_0^a$$

$$= \frac{2}{\sqrt{3a}} \left(\frac{2a^{5/2}}{5} - \frac{2a^{3/2}}{3} - 0 - 0 \right) = \frac{4}{\sqrt{3a}} \left(\frac{a^{5/2}}{5} - \frac{a^{3/2}}{3} \right)$$

$$= \frac{4a^{5/2}}{\sqrt{3a}} \left(\frac{1}{5} - \frac{1}{3} \right) = \frac{4a^{5/2}}{\sqrt{3a}} \left(-\frac{2}{15} \right)$$

$$= -\frac{8a^{5/2}}{15\sqrt{3a}} = -\frac{8a^2}{15\sqrt{3}}$$

But Area is always +ve $\therefore \text{Area} = \frac{8a^2}{15\sqrt{3}}$

Q8.

$$x^2(x^2+y^2) = a^2(y^2-x^2)$$

Put $y = -y$

No change So curve is symmetric about x -axis

Put $x = -x$

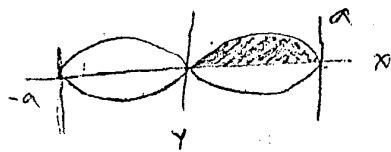
No change So curve is symmetric about y -axis

Asymptote $\parallel$ to y -axis $\{x = (a)^{1/2}\}$

$$x^4 + x^2y^2 - a^2y^2 + a^2x^2 = 0$$

$$x^4 + x^2(x^2 - a^2) + a^2x^2 = 0$$

7.5-6



Coast of highest power $y=0$

$$x^2 - a^2 = 0$$

$$x = \pm a$$

$$x^4 + x^2 y^2 - a^2 y^2 + a^2 x = 0$$

$$y^2(x^2 - a^2) = -x^4 - a^2 x$$

$$y^2 = \frac{-x^4 - a^2 x}{(x^2 - a^2)}$$

$$y = \frac{x\sqrt{x^2 + a^2}}{\sqrt{a^2 - x^2}}$$

$$x \div \text{by } \sqrt{a^2 + x^2}$$

$$\text{Area} = \int_a^b y dx = 4 \int_0^a \frac{x\sqrt{x^2 + a^2}}{\sqrt{a^2 - x^2}} dx$$

$$= 4 \int_0^a \frac{x\sqrt{x^2 + a^2}}{\sqrt{a^2 - x^2}} \cdot \frac{\sqrt{a^2 + x^2}}{\sqrt{a^2 + x^2}} dx$$

$$= 4 \int_0^a \frac{x(x^2 + a^2)}{\sqrt{a^4 - x^4}} dx$$

$$= \int_0^a \frac{x^3 dx}{\sqrt{a^4 - x^4}} + 4 \int_0^a \frac{xa^2}{\sqrt{a^4 - x^4}} dx$$

$$= -\int_0^a (a^4 - x^4)^{-\frac{1}{2}} \frac{1}{4} x^3 dx + 2 \int_0^a \frac{a^2 dz}{\sqrt{a^4 - z^2}}$$

$$= -\left| \frac{(a^4 - x^4)^{\frac{1}{2}}}{\frac{1}{2}} \right|_0^a + 2a^2 \int_0^a \frac{dz}{\sqrt{a^4 - z^2}}$$

$$= -\left[2\left\{ (a^4 - x^4)^{\frac{1}{2}} - (a^4 - a^4)^{\frac{1}{2}} \right\} \right] + 2a^2 \left[\sin^{-1} \frac{z}{a^2} \right]_0^a$$

$$= -2\{0 - a^2\} + 2a^2 \left(\sin^{-1} \left(\frac{a^2}{a^2} \right) - \sin^{-1}(0) \right) \text{ by conditions given}$$

$$A = 2a^2 + 2a^2 \left(\frac{\pi}{2} - 0 \right)$$

$$A = 2a^2 + 2a^2 \frac{\pi}{2} = a^2(2 + \pi) \text{ Ans.}$$

Also because $x^2 = a^2 \cos \theta$ or $x^2 = a^2 \sin \theta$
 $dx = -a^2 \sin \theta d\theta$
 $\theta = 0, \pi/2$

Put $x^2 = z$
 $2x dx = dz$
 when $x=0, z=0$
 when $x=a, z=a^2$

Q.9. $xy^2 = 4(2-x)$

$$y^2 = \frac{4(2-x)}{x} \Rightarrow y = \pm \frac{2\sqrt{2-x}}{\sqrt{x}}$$

The curve is symmetric about x -axis ^{No change.}
(y by $-y$)

Put $y=0$, $\frac{2\sqrt{2-x}}{\sqrt{x}} = 0 \Rightarrow \sqrt{2-x} = 0$

$$\Rightarrow x=2$$

Asymptote // to y -axis $x=0$

$y = \pm 2 \frac{\sqrt{2-x}}{\sqrt{x}}$ Required Area = 2 (Shaded Area)

$$A = 2 \int_0^2 \frac{2\sqrt{2-x}}{\sqrt{x}} dx$$

Put $x = 2 \sin^2 \theta$

$$dx = 4 \sin \theta \cos \theta d\theta$$

$$= 4 \int_0^{\pi/2} \frac{\sqrt{2-2\sin^2 \theta}}{\sqrt{2\sin^2 \theta}} 4 \sin \theta \cos \theta d\theta$$

when $x=0$ $\theta=0$

when $x=2$ $\theta=\frac{\pi}{2}$

$$= 16 \int_0^{\pi/2} \frac{\cos \theta}{\sin \theta} (\sin \theta \cos \theta) d\theta$$

$$= 16 \int_0^{\pi/2} \cos^2 \theta d\theta = 16 \int_0^{\pi/2} \left(\frac{1+\cos 2\theta}{2} \right) d\theta$$

$$= 8 \int_0^{\pi/2} d\theta + 8 \int_0^{\pi/2} \cos 2\theta d\theta$$

$$= 8 \left[\theta \right]_0^{\pi/2} + 8 \left[\frac{\sin 2\theta}{2} \right]_0^{\pi/2}$$

$$= 8 \left(\frac{\pi}{2} \right) + 8 \left(\frac{\sin \pi}{2} - \frac{\sin 0}{2} \right)$$

$$= 4\pi - 0 = 4\pi \text{ Ans.}$$

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7.5-8

Q10. $x^2 y^2 = a^2 (y^2 - x^2)$

Asymptotes // to y -axis

$$x^2 y^2 = a^2 y^2 + a^2 x^2 = 0$$

$$y^2 (x^2 - a^2) + a^2 x^2 = 0$$

$$x^2 - a^2 = 0 \Rightarrow x = \pm a$$

The curve is symmetric about x -axis & y -axis ($\because$ even powers of x & y)

y is real for $-a \leq x \leq a$

Required area = $4 \int_0^a \frac{ay}{\sqrt{a^2 - x^2}} dx$

$$y^2 (x^2 - a^2) + a^2 x^2 = 0$$

$$y^2 = \frac{-a^2 x^2}{(x^2 - a^2)}$$

$$= \frac{a^2 x^2}{(a^2 - x^2)}$$

$$= -4a \int_0^a (a^2 - x^2)^{-\frac{1}{2}} (-x dx)$$

$$= -2a \int_0^a (a^2 - x^2)^{-\frac{1}{2}} (-2x dx)$$

$$y = \pm \frac{ax}{\sqrt{a^2 - x^2}}$$

$$= -2a \left[\frac{(a^2 - x^2)^{\frac{1}{2}}}{\frac{1}{2}} \right]_0^a$$

$$= -4a \left[(a^2 - x^2)^{\frac{1}{2}} - (a^2 - 0)^{\frac{1}{2}} \right] = -4a \cdot (0 - (a^2)^{\frac{1}{2}})$$

$$= 4a^2 \text{ Ans.}$$

Q11. $ay^2 = x^2(a-x)$

The curve is symmetric only about x -axis ($\because$ even power of y)

Put $y=0 \Rightarrow x^2(a-x)=0 \Rightarrow x=0, a$

Area = $2 \int_0^a \frac{x\sqrt{a-x}}{\sqrt{a}} dx$

$$y^2 = \frac{x^2(a-x)}{a}$$

$$y = \pm \frac{x\sqrt{a-x}}{\sqrt{a}}$$

7.5-9

Put $x = a \sin^2 \theta$ - $dx = 2a \sin \theta \cos \theta d\theta$

when $x=0$ - $\sin^2 \theta = 0$ - $\theta = 0$

when $x=a$ - $\sin^2 \theta = 1$ - $\theta = 90$

$$\text{Area} = 2 \int_0^{\pi/2} \frac{a \sin^2 \theta \sqrt{a - a \sin^2 \theta}}{\sqrt{a}} \times 2a \sin \theta \cos \theta d\theta$$

$$= 4a^2 \int_0^{\pi/2} \frac{\sin^2 \theta \sqrt{1 - \sin^2 \theta} \sin \theta \cos \theta d\theta}{\sqrt{a}}$$

$$= 4a^2 \int_0^{\pi/2} \sin^3 \theta \cos^2 \theta d\theta$$

$$= 4a^2 \int_0^{\pi/2} \sin^3 \theta \cos^2 \theta d\theta$$

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Using Walli's Formula $\int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{(p-1)(p-3) \dots (2-1)(1-1)}{(p+q)(p+q-2) \dots (p+q-4)}$

$$= 4a^2 \int_0^{\pi/2} \sin^3 \theta \cos^2 \theta d\theta = 4a^2 \left(\frac{(3-1) \dots (2-1)}{(3+2)(3+2-2)(3+2-4)} \right)$$

$$4a^2 \left(\frac{2}{(5 \times 3)(1)} \right) = \frac{8a^2}{15}$$

Ans: $a^2 x^2 = y^3 (2a - y)$

Change y by $-y$, Curve is changed. So not symmetric about x -axis.

Change x by $-x$, Curve is not changed, So symmetric about y -axis.

For y -Intercept Put $x=0$ - $y^3 (2a - y) = 0$
 $y = 0, 2a$

7.5-16

$$\text{Area} = \int_a^b x \, dy$$

Required Area = 2 (Shaded Portion)

$$A = 2 \int_0^{2a} \frac{y^{3/2}}{a} \sqrt{(2a-y)} \, dy$$

$$\text{Put } y = 2a \sin^2 \theta$$

$$dy = 4a \sin \theta \cos \theta \, d\theta$$

$$\text{when } y=0 \quad 0 = \sin^2 \theta \Rightarrow \theta = 0$$

$$\text{when } y=2a \quad 1 = \sin^2 \theta \Rightarrow \theta = \pi/2$$

$$A = 2 \int_0^{\pi/2} \frac{(2a \sin^2 \theta)^{3/2}}{a} \sqrt{2a - 2a \sin^2 \theta} \{4a \sin \theta \cos \theta \, d\theta\}$$

$$= \frac{2}{a} \int_0^{\pi/2} 2^{3/2} a^{3/2} \sin^3 \theta \cdot 2a \sqrt{1 - \sin^2 \theta} (4 \sin \theta \cos \theta \, d\theta)$$

$$= 32a^2 \int_0^{\pi/2} \sin^4 \theta \cos^2 \theta \, d\theta$$

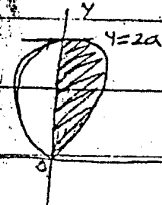
$$= 32a^2 \int_0^{\pi/2} \sin^4 \theta \cos^2 \theta \, d\theta$$

$$= 32a^2 \left(\frac{(4-1)(4-3) \dots (2-1)}{(4+2)(4+2-2)(4+2-4) \dots} \right) \frac{\pi}{2}$$

$$= 32a^2 \left(\frac{3 \cdot 1 \cdot 1}{6 \cdot 4 \cdot 2} \right) \frac{\pi}{2} = \frac{32a^2(3)}{6 \cdot 4 \cdot 2} \frac{\pi}{2} = \frac{96a^2\pi}{96} = a^2\pi$$

$a^2\pi$ is the area of circular disc of radius a .

Curve passes through origin



{ Shaded portion is bounded by the curve, y-axis, y=0, y=2a. }

$$a^2 x^2 = y^3 (2a-y)$$

$$x^2 = \frac{y^3}{a^2} (2a-y)$$

$$x = \pm \frac{y^{3/2}}{a} \sqrt{2a-y}$$

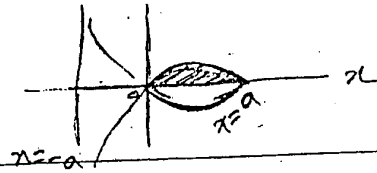
x is real for $0 \leq y \leq 2a$

Wallis Formula.

$$\int_0^{\pi/2} \sin^p x \cos^q x \, dx = \frac{(p-1)(p-3) \dots (1-1)(1-3) \dots}{(p+1)(p+1-2)(p+1-4) \dots} \frac{\pi}{2}$$

(1.1)

7.5-11



Q13. Area of the region bounded by the loop of the curve $y^2(a-x) = x^2(a-x)$

The curve is symmetric about x-axis. (Even power of y)

x-Intercept Put $y=0$ $x^2(a-x)=0$ $x=0, a$

$$\text{So Area} = 2 \int_0^a x \frac{\sqrt{a-x}}{\sqrt{a+x}} dx$$

$$y^2 = \frac{x^2(a-x)}{a+x}$$

$$y = x \sqrt{\frac{a-x}{a+x}}$$

Put $x = a \cos \theta$

$$dx = -a \sin \theta d\theta$$

$$x=0 \rightarrow \cos \theta = 0 \quad \theta = \frac{\pi}{2}$$

$$x=a \rightarrow \cos \theta = \frac{a}{a} = 1 \quad \theta = 0$$

$$\text{Area} = 2 \int_{\pi/2}^0 a \cos \theta \sqrt{\frac{1-\cos \theta}{1+\cos \theta}} (-a \sin \theta) d\theta$$

$$= -2 \int_{\pi/2}^0 a^2 \cos \theta \sin \theta \left(\frac{\sqrt{1-\cos \theta}}{\sqrt{1+\cos \theta}} \right) d\theta$$

$$= -2a^2 \int_{\pi/2}^0 (1-2\sin^2 \frac{\theta}{2}) (2\sin \frac{\theta}{2} \cos \frac{\theta}{2}) \left(\frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} \right) d\theta$$

$$= -2a^2 \int_{\pi/2}^0 (2\sin^2 \frac{\theta}{2} - 1, \sin^4 \frac{\theta}{2}) d\theta$$

$$= -4a^2 \int_{\pi/2}^0 \left(\sin^2 \frac{\theta}{2} - 2\sin^4 \frac{\theta}{2} \right) d\theta = -4a^2 \int_{\pi/2}^0 \sin^2 \frac{\theta}{2} d\theta + 8a^2 \int_{\pi/2}^0 \sin^4 \frac{\theta}{2} d\theta$$

$$= +4a^2 \int_0^{\pi/2} \left(\sin^2 \frac{\theta}{2} d\theta - 8a^2 \int_0^{\pi/2} \sin^4 \frac{\theta}{2} d\theta \right)$$

$$\cos \theta = 2\cos^2 \frac{\theta}{2} - 1$$

$$1 + \cos \theta = 2\cos^2 \frac{\theta}{2}$$

$$\cos \theta = 1 - 2\sin^2 \frac{\theta}{2}$$

$$1 - \cos \theta = 2\sin^2 \frac{\theta}{2}$$

7.5-12

$$\begin{aligned}
 & 4a^2 \int_0^{\pi/2} \left(\frac{1 - \cos \theta}{2} \right) d\theta - 8a^2 \int_0^{\pi/2} \left(\frac{1 - \cos \theta}{2} \right) d\theta \\
 &= \frac{4a^2}{2} \left[\theta - \sin \theta \right]_0^{\pi/2} - 8a^2 \int_0^{\pi/2} \left(\frac{1 + \cos^2 \theta - 2\cos \theta}{4} \right) d\theta \\
 &= 2a^2 \left(\frac{\pi}{2} - \sin \frac{\pi}{2} - 0 - 0 \right) - 2a^2 \int_0^{\pi/2} \left(1 + \frac{1 + \cos 2\theta}{2} - 2\cos \theta \right) d\theta \\
 &= 2a^2 \frac{\pi}{2} - 2a^2 \sin \left(\frac{\pi}{2} \right) - \frac{2a^2}{2} \int_0^{\pi/2} (2 + 1 + \cos 2\theta - 4\cos \theta) d\theta \\
 &= a^2 \pi - 2a^2 (1) - a^2 \int_0^{\pi/2} (3 + \cos 2\theta - 4\cos \theta) d\theta \\
 &= a^2 \pi - 2a^2 - a^2 \left[3\theta + \frac{\sin 2\theta}{2} - 4\sin \theta \right]_0^{\pi/2} \\
 &= a^2 \pi - 2a^2 - a^2 \left[3\frac{\pi}{2} + 0 - 4 \right] \\
 &= a^2 \pi - 2a^2 - 3a^2 \frac{\pi}{2} + 4a^2 = \frac{2a^2 \pi - 4a^2 - 3a^2 \pi + 8a^2}{2} \\
 &= \frac{-a^2 \pi + 4a^2}{2} = \boxed{\frac{a^2}{2} (4 - \pi)}
 \end{aligned}$$

Now area between the curve & asymptote

Asymptote // to y-axis

$$x + a = 0$$

$$x = -a$$

$$\text{Area} = 2 \int_{-a}^0 \frac{x \sqrt{a-x}}{a-x} dx$$

$$\text{Put } x = a \cos \theta \quad dx = -a \sin \theta d\theta$$

$$x = -a, \theta = \pi \quad x = 0, \theta = \frac{\pi}{2}$$

$$= 2 \int_{\pi/2}^{\pi} \frac{a \cos \theta \sqrt{a - a \cos \theta}}{a + a \cos \theta} (-a \sin \theta d\theta)$$

$$= 2a^2 \int_{\pi/2}^{\pi} \frac{\cos \theta \sin \theta \sqrt{1 - \cos \theta}}{1 + \cos \theta} d\theta = 2a^2 \left[\theta - \sin \theta \right]_{\pi/2}^{\pi} - a^2 \left[3\theta + \frac{\sin 2\theta}{2} - 4\sin \theta \right]_{\pi/2}^{\pi}$$

7.5-13

$$= 2a^2 \left[\pi - \sin \pi - \frac{\pi}{2} + \frac{\sin \frac{\pi}{2}}{2} \right] - a^2 \left(3\pi + \frac{\sin 2\pi}{2} - 4 \sin \pi - \frac{3\pi}{2} - \frac{\sin^2 \frac{\pi}{2}}{2} + 4 \sin^2 \frac{\pi}{2} \right)$$

$$= 2a^2 \left(\pi - 0 - \frac{\pi}{2} + 1 \right) - a^2 \left(3\pi + 0 - 0 - \frac{3\pi}{2} - 0 + 4 \right)$$

$$= 2a^2 \left(\frac{\pi}{2} + 1 \right) - a^2 \left(3\pi - \frac{3\pi}{2} + 4 \right)$$

$$= \frac{2a^2 \pi}{2} + 2a^2 - a^2 \left(\frac{3\pi}{2} + 4 \right)$$

$$= a^2 \pi + 2a^2 - \frac{a^2 3\pi}{2} - 4a^2$$

$$= \frac{2a^2 \pi + 4a^2 - a^2 3\pi - 8a^2}{2}$$

$$= \frac{-a^2 \pi - 4a^2}{2}$$

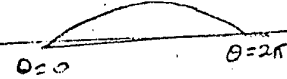
$$\text{Area} = \frac{-a^2(\pi + 4)}{2}$$

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CMU $x = a(1 - \sin \theta)$ $y = a(1 - \cos \theta)$

$dx = a(1 - \cos \theta) d\theta$

$$A = \int_0^{2\pi} a(1 - \cos \theta) a(1 - \cos \theta) d\theta$$



$$= a^2 \int_0^{2\pi} (1 - \cos \theta)^2 d\theta = a^2 \int_0^{2\pi} (1 + \cos^2 \theta - 2\cos \theta) d\theta$$

$$= a^2 \int_0^{2\pi} \left(1 + \frac{1 + \cos 2\theta}{2} - 2\cos \theta \right) d\theta$$

$$= a^2 \int_0^{2\pi} \left(\frac{2+1+\cos 2\theta}{2} - 2\cos \theta \right) d\theta$$

7.5-14

$$\frac{a^2}{2} \int_0^{2\pi} (3 + \cos 2\theta - 4 \cos \theta) d\theta$$

$$= \frac{a^2}{2} \left[3\theta + \frac{\sin 2\theta}{2} - 4 \sin \theta \right]_0^{2\pi}$$

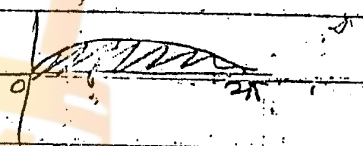
$$= \frac{a^2}{2} \left[3(2\pi) + \frac{\sin 2(2\pi)}{2} - 4 \sin(2\pi) - 0 + 0 - 0 \right]$$

$$= \frac{a^2}{2} 6\pi = \boxed{3a^2\pi}$$

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Q15 $x = a(\theta + \sin \theta)$

$y = a(1 - \cos \theta)$



$dx = a(1 + \cos \theta) d\theta$

$$\int_0^{2\pi} y dx = \int_0^{2\pi} a(1 - \cos \theta) a(1 + \cos \theta) d\theta$$

$$= a^2 \int_0^{2\pi} (1 - \cos^2 \theta) d\theta = a^2 \int_0^{2\pi} \sin^2 \theta d\theta$$

$$= a^2 \int_0^{2\pi} \left(\frac{1 - \cos 2\theta}{2} \right) d\theta = \frac{a^2}{2} \int_0^{2\pi} d\theta - \frac{a^2}{2} \int_0^{2\pi} \cos 2\theta d\theta$$

$$= \frac{a^2}{2} \left[\theta \right]_0^{2\pi} - \frac{a^2}{2} \left[\frac{\sin 2\theta}{2} \right]_0^{2\pi}$$

$$= \frac{a^2}{2} (2\pi) - \frac{a^2}{2} \left(\frac{\sin 2(2\pi)}{2} - 0 \right)$$

Area = $a^2\pi - 0 = a^2\pi$

Q15 Area of the region bounded by the loop of the curve
 $x^4 + y^4 = 2a^2 xy$.

Put $x = r \cos \theta$, $y = r \sin \theta$.

$$r^4 \cos^4 \theta + r^4 \sin^4 \theta = 2a^2 r \cos \theta r \sin \theta$$

$$r^4 (\cos^4 \theta + \sin^4 \theta) = 2a^2 r^2 \cos \theta \sin \theta$$

$$\frac{r^4}{r^2} = \frac{2a^2 \sin \theta \cos \theta}{\cos^4 \theta + \sin^4 \theta}$$

$$r^2 = \frac{2a^2 \sin \theta \cos \theta}{\cos^4 \theta + \sin^4 \theta}$$

Put $r=0 \Rightarrow 2a^2 \sin \theta \cos \theta = 0$

$$\sin \theta \cos \theta = 0$$

$$\sin \theta = 0$$

$$\theta = 0$$

$$\cos \theta = 0$$

$$\theta = \pi/2$$

Area = $\frac{1}{2} \int_0^{\pi/2} r^2 d\theta$

$$= \frac{1}{2} \int_0^{\pi/2} \frac{2a^2 \sin \theta \cos \theta}{\cos^4 \theta + \sin^4 \theta} d\theta$$

$$= \frac{2a^2}{2} \int_0^{\pi/2} \frac{\sin \theta \cos \theta}{\cos^4 \theta + \sin^4 \theta} d\theta$$

Divide N & D by $\cos^4 \theta$

$$= a^2 \int_0^{\pi/2} \frac{\tan \theta \sec^2 \theta}{1 + \tan^4 \theta} d\theta$$

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7.5-16

Put $\tan \theta = u$.

$2 \tan \theta \sec^2 \theta d\theta = du$.

when $\theta = 0$ $u = 0$

when $\theta = \frac{\pi}{2}$ $u = \infty$

$\tan \theta \sec^2 \theta d\theta = \frac{du}{2}$

Area = $a^2 \int_0^{\infty} \frac{du/2}{1+u^2}$

$= \frac{a^2}{2} \int_0^{\infty} \frac{du}{1+u^2} = \frac{a^2}{2} \left[\tan^{-1} u \right]_0^{\infty}$

$= \frac{a^2}{2} \left[\tan^{-1}(\infty) - \tan^{-1}(0) \right]$

$= \frac{a^2}{2} \left(\frac{\pi}{2} - 0 \right) = \frac{\pi a^2}{4}$

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Q16. $(x+y)^2(x^2+y^2) = 2xy$

Put $x = r \cos \theta$

$y = r \sin \theta$

$(r \cos \theta + r \sin \theta)^2 (r^2 \cos^2 \theta + r^2 \sin^2 \theta) = 2r^2 \cos \theta r \sin \theta$

$(r^2 \cos^2 \theta + r^2 \sin^2 \theta + 2r^2 \sin \theta \cos \theta) (r^2 (\cos^2 \theta + \sin^2 \theta)) = 2r^3 \cos \theta \sin \theta$

$r^2 (\cos^2 \theta + \sin^2 \theta + 2 \sin \theta \cos \theta) = 2r^2 \cos \theta \sin \theta$

$r^2 (1 + 2 \sin \theta \cos \theta) = 2r^2 \sin \theta \cos \theta$

$r^2 = \frac{2r^2 \sin \theta \cos \theta}{1 + 2 \sin \theta \cos \theta}$

Put $r=0 \Rightarrow 2r^2 \sin \theta \cos \theta = 0$

7.5-17

$$\sin \theta \cos \theta = 0$$

$$\sin \theta = 0$$

$$\cos \theta = 0$$

$$\theta = 0$$

$$\theta = \frac{\pi}{2}$$

$$\text{Area} = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta = \frac{1}{2} \int_0^{\pi/2} \frac{a^2 2 \sin \theta \cos \theta}{1 + 2 \sin \theta \cos \theta} d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi/2} \frac{\sin 2\theta}{1 + \sin 2\theta} d\theta = \frac{a^2}{2} \int_0^{\pi/2} \frac{(\sin 2\theta + 1) - 1}{\sin 2\theta + 1} d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi/2} \left(1 - \frac{1}{\sin 2\theta + 1} \right) d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi/2} d\theta - \frac{a^2}{2} \int_0^{\pi/2} \frac{d\theta}{1 + \sin 2\theta}$$

$$= \frac{a^2}{2} \left| \theta \right|_0^{\pi/2} - \frac{a^2}{2} \int_0^{\pi/2} \frac{d\theta}{1 + \cos(90 - 2\theta)}$$

$$= \frac{a^2}{2} \left(\frac{\pi}{2} - 0 \right) - \frac{a^2}{2} \int_0^{\pi/2} \frac{d\theta}{2 \cos^2(90 - 2\theta)}$$

$$= \frac{a^2 \pi}{4} - \frac{a^2}{2} \int_0^{\pi/2} \frac{d\theta}{2 \cos^2(45 - \theta)}$$

$$= \frac{a^2 \pi}{4} - \frac{a^2}{4} \int_0^{\pi/2} \sec^2(45 - \theta) d\theta$$

$$= \frac{a^2 \pi}{4} - \frac{a^2}{4} \left| \frac{\tan(45 - \theta)}{-1} \right|_0^{\pi/2}$$

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$$\sin \theta = \cos(90 - \theta)$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$1 + \cos 2\theta = 2 \cos^2 \theta$$

$$\int \sec^2 \theta d\theta = \tan \theta$$

7.5-18

$$= \frac{a^2 \pi}{4} + \frac{a^2}{4} \left(\tan\left(\frac{\pi}{4} - \frac{\pi}{2}\right) - \tan\left(\frac{\pi}{4} - 0\right) \right)$$

$$= \frac{a^2 \pi}{4} + \frac{a^2}{4} \left(\tan\left(-\frac{\pi}{4}\right) - \tan\left(\frac{\pi}{4}\right) \right)$$

$$= \frac{a^2 \pi}{4} + \frac{a^2}{4} \left(-\tan\left(\frac{\pi}{4}\right) - \tan\left(\frac{\pi}{4}\right) \right)$$

$$= \frac{a^2 \pi}{4} + \frac{a^2}{4} (-2 \tan \frac{\pi}{4})$$

$$\tan \frac{\pi}{4} = 1$$

$$= \frac{a^2 \pi}{4} - \frac{a^2}{2} = \frac{a^2}{2} \left(\frac{\pi}{2} - 1 \right)$$

Q17

Area of the region bounded by the circle $r=a$ & $r=a \cos 5\theta$

Area of the circle $= \pi a^2$

Symmetrical about initial line ($\theta=0$)

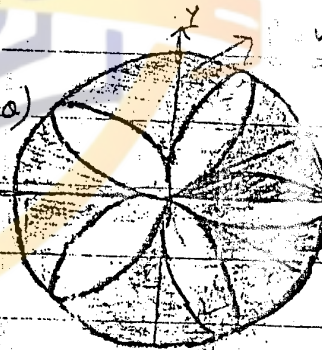
It has 5 loops.

For limits of loops $r=0$.

$$0 = a \cos 5\theta$$

$$\cos 5\theta = 0 \Rightarrow 5\theta = \frac{\pi}{2}$$

$$\theta = \frac{\pi}{10}$$



Area

$$\theta = 0$$

$$\theta = \frac{\pi}{10}$$

$$\text{Area of one loop} = 2 \int_0^{\frac{\pi}{10}} \frac{r^2}{2} d\theta = \frac{1}{2} \int_0^{\frac{\pi}{10}} a^2 \cos^2 5\theta d\theta$$

7.5-19

$$= a^2 \int_0^{\pi/10} \cos^2 5\theta \, d\theta$$

$$= a^2 \int_0^{\pi/10} \left(\frac{1 + \cos 10\theta}{2} \right) d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi/10} d\theta + \frac{a^2}{2} \int_0^{\pi/10} \cos 10\theta \, d\theta$$

$$= \frac{a^2}{2} \left| \theta \right|_0^{\pi/10} + \frac{a^2}{2} \left| \frac{\sin 10\theta}{10} \right|_0^{\pi/10}$$

$$= \frac{a^2}{2} \left(\frac{\pi}{10} \right) + \frac{a^2}{20} \left(\sin 10 \left(\frac{\pi}{10} \right) - \sin 10(0) \right)$$

$$\text{Area of one loop} = \frac{a^2 \pi}{20} + 0 - 0$$

$$\text{Area of 5 loops} = 5 \times \frac{a^2 \pi}{20} = \frac{a^2 \pi}{4}$$

$$\text{Area of the circle} = \pi a^2$$

Area bounded by $r = a$ & $r = a \cos 5\theta$ is

$$\pi a^2 - \frac{a^2 \pi}{4} = \frac{3\pi a^2}{4} \quad \text{Ans.}$$

Q18. $r = a(1 + \cos \theta)$

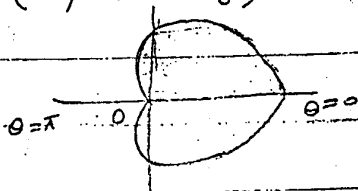
It is symmetrical about initial line ($\theta = 0$, No change)

Put $r = 0$

$$a(1 + \cos \theta) = 0$$

$$1 + \cos \theta = 0$$

$$\cos \theta = -1 \Rightarrow \theta = \pi$$



7.5-20

$$\text{Area of Cardiod} = \frac{1}{2} \int_0^{\pi} r^2 d\theta = \frac{2 \cdot 1}{2} \int_0^{\pi} a^2 (1 + \cos \theta)^2 d\theta$$

$$= a^2 \int_0^{\pi} (2 \cos^2 \frac{\theta}{2})^2 d\theta$$

$$\cos \theta = 2 \cos^2 \frac{\theta}{2} - 1$$

$$= 4a^2 \int_0^{\pi} \cos^4 \frac{\theta}{2} d\theta$$

$$\text{Put } \frac{\theta}{2} = u$$

$$= 4a^2 \int_0^{\frac{\pi}{2}} \cos^4 u \cdot 2du$$

$$\frac{d\theta}{2} = du$$

$$d\theta = 2du$$

$$= 8a^2 \int_0^{\frac{\pi}{2}} \cos^4 u du$$

$$\text{when } \theta = 0 \quad u = 0$$

$$\text{when } \theta = \pi \quad u = \frac{\pi}{2}$$

Using Wallis' Formula

$$= 8a^2 \int_0^{\frac{\pi}{2}} \cos^4 u du = 8a^2 \left(\frac{4-1}{4} \cdot \frac{4-3}{4-2} \cdot \frac{\pi}{2} \right)$$

$$\int_0^{\frac{\pi}{2}} \cos^n x dx = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{1}{2} \cdot \frac{\pi}{2}$$

$$= 8a^2 \left(\frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \right) = \frac{3(3a^2\pi)}{6} = \frac{3a^2\pi}{2} \text{ Ans}$$

Q19 $r^2 = a^2 \cos 2\theta$

It is symmetrical about initial line (θ by $-\theta$, unchanged)

It is symmetrical about y-axis (θ by $(\pi - \theta)$, unchanged)

Put $r = 0$ $a^2 \cos 2\theta = 0$

$$\cos 2\theta = 0$$

$$2\theta = \cos^{-1}(0) = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{4}$$

7.5-21

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$$= \frac{9a^2}{2} \int_0^{\frac{\pi}{2}} \frac{\tan^2 \theta \sec^2 \theta}{(1 + \tan^3 \theta)^2} d\theta$$

$$= \frac{3a^2}{2} \int_0^{\frac{\pi}{2}} (1 + \tan^3 \theta)^{-2} \cdot 3 \tan^2 \theta \sec^2 \theta d\theta$$

$$= \frac{3a^2}{2} \left| \frac{(1 + \tan^3 \theta)^{-1}}{-1} \right|_0^{\frac{\pi}{2}} = -\frac{3a^2}{2} \left| \frac{1}{1 + \tan^3 \theta} \right|_0^{\frac{\pi}{2}}$$

$$= -\frac{3a^2}{2} \left[\frac{1}{1 + \tan^3(\frac{\pi}{2})} - \frac{1}{1 + 0} \right]$$

$$= -\frac{3a^2}{2} \left[\frac{1}{\infty} - 1 \right] = -\frac{3a^2}{2} [0 - 1]$$

$$= \frac{3a^2}{2} \text{ Ans.}$$

Imp
Q21

$$r = \frac{6}{2 - \cos \theta}$$

①

$$r = 0 \Rightarrow \frac{6}{2 - \cos \theta} = 0$$

6 = 0 impossible

∴ change in Cartesian form

Change into Cartesian Coord System

$$r = \sqrt{x^2 + y^2} \quad x = r \cos \theta$$

$$\therefore \textcircled{1} \text{ becomes } 2r - r \cos \theta = 6$$

$$2\sqrt{x^2 + y^2} - x = 6$$

$$2\sqrt{x^2 + y^2} = 6 + x$$

$$\text{Squaring } 4(x^2 + y^2) = x^2 + 36 + 12x$$

7.5-22

$$\begin{aligned}
 \text{Area} &= 4 \int_0^{\frac{\pi}{4}} \frac{r^2 d\theta}{2} = \frac{4}{2} \int_0^{\frac{\pi}{4}} a^2 \cos 2\theta d\theta \\
 &= 2a^2 \int_0^{\frac{\pi}{4}} \cos 2\theta d\theta \\
 &= 2a^2 \left[\frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{4}} \\
 &= \frac{2a^2}{2} \left(\sin\left(\frac{\pi}{2}\right) - \sin 2(0) \right) = a^2 \left(\sin\left(\frac{\pi}{2}\right) \right)
 \end{aligned}$$

Area = a^2 .

Q20. $r = \frac{3a \sin \theta \cos \theta}{\cos^3 \theta + \sin^3 \theta}$ folium $x^3 + y^3 = 3axy$

Put $r=0$

$$3a \sin \theta \cos \theta = 0$$

$$\sin \theta \cos \theta = 0$$

$$\sin \theta = 0$$

$$\theta = 0$$

$$\cos \theta = 0$$

$$\theta = \frac{\pi}{2}$$

$$\begin{aligned}
 \text{Area} &= \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta = \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{(3a \sin \theta \cos \theta)^2}{(\cos^3 \theta + \sin^3 \theta)^2} d\theta
 \end{aligned}$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{9a^2 \sin^2 \theta \cos^2 \theta}{(\cos^3 \theta + \sin^3 \theta)^2} d\theta$$

$$= \frac{9a^2}{2} \int_0^{\frac{\pi}{2}} \frac{\sin^2 \theta \cos^2 \theta}{(\cos^3 \theta + \sin^3 \theta)^2} d\theta$$

$$= \frac{9a^2}{2} \int_0^{\frac{\pi}{2}} \frac{\sin^2 \theta \cos^2 \theta \cdot \frac{1}{\cos^4 \theta}}{\left(\frac{\cos^3 \theta}{\cos^4 \theta} + \frac{\sin^3 \theta}{\cos^4 \theta} \right)^2} d\theta$$

7.5-23

$$4x^2 + 4y^2 - 12x - 4y^2 = 36$$

$$3x^2 - 12x + 4y^2 = 36$$

$$3(x^2 - 4x) + 4y^2 = 36$$

$$3(x^2 - 4x + 4 - 4) + 4y^2 = 36$$

Completing Square

$$3((x-2)^2 - 4) + 4y^2 = 36$$

$$3(x-2)^2 - 12 + 4y^2 = 36$$

$$3(x-2)^2 + 4y^2 = 48$$

$$3(x-2)^2 + 4y^2 = 1$$

$$\frac{48}{48} \quad \frac{48}{48}$$

$$\frac{(x-2)^2}{16} + \frac{y^2}{12} = 1$$

$$\frac{(x-2)^2}{(4)^2} + \frac{y^2}{(\sqrt{12})^2} = 1$$

$$\text{Area of ellipse } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

is $ab\pi$.

$$a=4 \quad b=\sqrt{12}$$

So Area of the curve is $4(\sqrt{12})\pi$

$$= 4(2\sqrt{3})\pi = 8\sqrt{3}\pi$$

Q22. Area of the region that lies

outside the cardioid $r_1 = 1 + \cos\theta$

and inside the circle $r_2 = 3\cos\theta$.

$$r_1 = 1 + \cos\theta$$

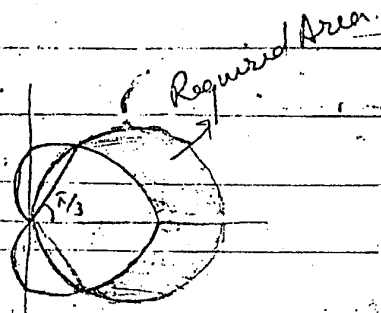
$$r_2 = 3\cos\theta$$

$$\therefore 1 + \cos\theta = 3\cos\theta$$

$$1 = 2\cos\theta$$

$$\frac{1}{2} = \cos\theta$$

$$\theta = \pm \frac{\pi}{3}$$



7.5-24

$$\text{Area} = \int_a^b \overset{\text{Cartesian}}{f(x) - g(x)} dx = \int_a^b \overset{\beta \text{ Polar}}{\frac{(r_1^2 - r_2^2)}{2}} d\theta$$

$$\text{Required Area} = 2 \int_0^{\pi/3} \frac{1}{2} ((3\cos\theta)^2 - (1+\cos\theta)^2) d\theta$$

$$= \int_0^{\pi/3} (9\cos^2\theta - 1 - \cos^2\theta - 2\cos\theta) d\theta$$

$$= \int_0^{\pi/3} (8\cos^2\theta - 2\cos\theta - 1) d\theta$$

$$= \int_0^{\pi/3} \left(8 \frac{(1+\cos 2\theta)}{2} - 2\cos\theta - 1 \right) d\theta$$

$2\cos^2\theta - 1 = \cos 2\theta$

$$= \int_0^{\pi/3} (4 + 4\cos 2\theta - 2\cos\theta - 1) d\theta$$

$$= \int_0^{\pi/3} (4\cos 2\theta - 2\cos\theta + 3) d\theta$$

$$= \int_0^{\pi/3} 4\cos 2\theta d\theta - \int_0^{\pi/3} 2\cos\theta d\theta + 3 \int_0^{\pi/3} d\theta$$

$$= \left| \frac{4\sin 2\theta}{2} \right|_0^{\pi/3} - \left| 2\sin\theta \right|_0^{\pi/3} + 3 \left| \theta \right|_0^{\pi/3}$$

$$= \left(2\sin^2\left(\frac{\pi}{3}\right) \right) - 2\sin\left(\frac{\pi}{3}\right) + \frac{3\pi}{3}$$

$$= 2 \frac{\sqrt{3}}{2} - 2 \frac{\sqrt{3}}{2} + \pi$$

$$= \pi \text{ Ans}$$

7.5-25

Q 23 Inside $r^2 = 2a^2 \cos 2\theta$ and outside $r = a$.

$$r^2 = 2a^2 \cos 2\theta$$

$$r = a$$

$$r^2 = a^2$$

$$a^2 = 2a^2 \cos 2\theta$$

$$\frac{a^2}{2a^2} = \cos 2\theta$$

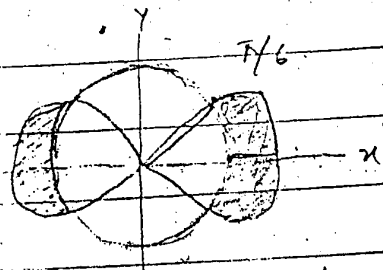
$$\Rightarrow \cos 2\theta = \frac{1}{2}$$

$$2\theta = \frac{\pi}{3}, -\frac{\pi}{3}$$

$$\theta = \frac{\pi}{6}, -\frac{\pi}{6}$$

Symmetric about initial line

Symmetric about y-axis



$$\text{Required Area} = 4 \int_{\pi/6}^{\pi/2} \frac{(r_1^2 - r_2^2)}{2} d\theta$$

$$= \frac{4}{2} \int_{\pi/6}^{\pi/2} (2a^2 \cos 2\theta - a^2) d\theta$$

$$= 2 \int_{\pi/6}^{\pi/2} 2a^2 \cos 2\theta d\theta - \int_{\pi/6}^{\pi/2} a^2 d\theta$$

$$= 4a^2 \left[\frac{\sin 2\theta}{2} \right]_{\pi/6}^{\pi/2} - 2a^2 \theta \Big|_{\pi/6}^{\pi/2}$$

$$= 2a^2 \left[\sin \left(2 \cdot \frac{\pi}{2} \right) - 0 \right] - 2a^2 \frac{\pi}{6}$$

$$= 2a^2 \left[\sin \frac{\pi}{3} \right] - a^2 \frac{\pi}{3}$$

$$= 2a^2 \left(\frac{\sqrt{3}}{2} \right) - a^2 \frac{\pi}{3}$$

$$= a^2 \sqrt{3} - a^2 \frac{\pi}{3}$$

$$= \frac{a^2}{3} (3\sqrt{3} - \pi) \text{ Ans.}$$

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$$7.5 - 26.1313$$

$$-41.48 = \theta$$

Q24 $r = 2 - 3\sin\theta$

Put $r = 0$

$$0 = 2 - 3\sin\theta \Rightarrow 3\sin\theta = 2$$

$$\theta = \sin^{-1} \frac{2}{3} = 41.48^\circ$$

The left half of smaller loop is determined by

$$\theta = \sin^{-1} \frac{2}{3} \text{ and } \theta = \frac{\pi}{2}$$

Area of smaller loop = 2 (shaded area) = $2 \int_{\sin^{-1}(\frac{2}{3})}^{\pi/2} \frac{r^2}{2} d\theta$

At 138.12° $r = 0$ again.

Symmetric about $\pi/2$
Replace θ by $(\pi - \theta)$ if needed

$$= \int_{\sin^{-1}(\frac{2}{3})}^{\pi/2} (2 - 3\sin\theta)^2 d\theta = \int_{\sin^{-1}(\frac{2}{3})}^{\pi/2} (4 + 9\sin^2\theta - 12\sin\theta) d\theta$$

$$= \int_{\sin^{-1}(\frac{2}{3})}^{\pi/2} \left(4 + 9 \left(\frac{1 - \cos 2\theta}{2} \right) - 12\sin\theta \right) d\theta = \int_{\sin^{-1}(\frac{2}{3})}^{\pi/2} \left(\frac{17}{2} - \frac{9}{2} \cos 2\theta - 12\sin\theta \right) d\theta$$

$$= \left| \frac{17}{2} \theta - \frac{9}{2} \frac{\sin 2\theta}{2} - 12(-\cos\theta) \right|_{\sin^{-1}(\frac{2}{3})}^{\pi/2} = \left| \frac{17}{2} \theta - \frac{9}{4} \sin 2\theta + 12\cos\theta \right|_{\sin^{-1}(\frac{2}{3})}^{\pi/2}$$

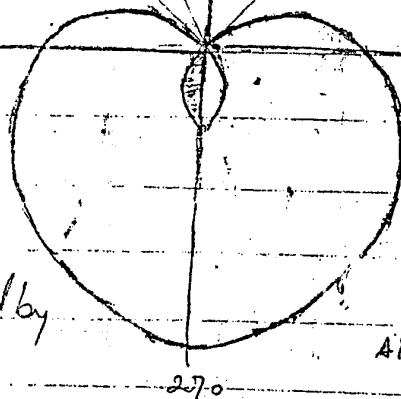
$$= \left| \frac{17}{2} \theta - \frac{9}{4} \sin\theta \cos\theta + 12\cos\theta \right|_{\sin^{-1}(\frac{2}{3})}^{\pi/2}$$

We know $\theta = \sin^{-1} \frac{2}{3} \Rightarrow \sin\theta = \frac{2}{3}$

Also $\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{4}{9}} = \sqrt{\frac{5}{9}} = \frac{\sqrt{5}}{3}$

$$= 2 \left(\frac{\sqrt{3}}{2} - 2 \frac{\sqrt{3}}{2} + \pi \right)$$

$$= \pi \text{ Ans.}$$



$$7.5 - 27$$

$$\begin{aligned}
 & \cos^{-1}\left(-\frac{2}{3}\right) \\
 & -2 \int_0^{\cos^{-1}\left(-\frac{2}{3}\right)} (2+3\cos\theta)^2 d\theta - 2 \int_{\cos^{-1}\left(-\frac{2}{3}\right)}^{\pi} (2+3\cos\theta)^2 d\theta \\
 & = \int_0^{\cos^{-1}\left(-\frac{2}{3}\right)} (4+9\cos^2\theta+12\cos\theta) d\theta + \int_{\cos^{-1}\left(-\frac{2}{3}\right)}^{\pi} (4+9\cos^2\theta+12\cos\theta) d\theta \\
 & = \int_0^{\cos^{-1}\left(-\frac{2}{3}\right)} \left(4+9\left(\frac{1+\cos 2\theta}{2}\right)+12\cos\theta\right) d\theta - \int_{\cos^{-1}\left(-\frac{2}{3}\right)}^{\pi} \left(4+9\left(\frac{1+\cos 2\theta}{2}\right)+12\cos\theta\right) d\theta \\
 & = \int_0^{\cos^{-1}\left(-\frac{2}{3}\right)} \left(4+\frac{9}{2}+\frac{9}{2}\cos 2\theta+12\cos\theta\right) d\theta - \int_{\cos^{-1}\left(-\frac{2}{3}\right)}^{\pi} \left(4+\frac{9}{2}+\frac{9}{2}\cos 2\theta+12\cos\theta\right) d\theta \\
 & = \int_0^{\cos^{-1}\left(-\frac{2}{3}\right)} \left(\frac{17}{2}+\frac{9}{2}\cos 2\theta+12\cos\theta\right) d\theta - \int_{\cos^{-1}\left(-\frac{2}{3}\right)}^{\pi} \left(\frac{17}{2}+\frac{9}{2}\cos 2\theta+12\cos\theta\right) d\theta \\
 & = \left[\frac{17\theta}{2}+\frac{9}{4}\sin 2\theta+12\sin\theta\right]_0^{\cos^{-1}\left(-\frac{2}{3}\right)} - \left[\frac{17\theta}{2}+\frac{9}{4}\sin 2\theta+12\sin\theta\right]_{\cos^{-1}\left(-\frac{2}{3}\right)}^{\pi} \\
 & = \left[\frac{17\theta}{2}+\frac{9}{4}2\sin\theta\cos\theta+12\sin\theta\right]_0^{\cos^{-1}\left(-\frac{2}{3}\right)} - \left[\frac{17\theta}{2}+\frac{9}{4}2\sin\theta\cos\theta+12\sin\theta\right]_{\cos^{-1}\left(-\frac{2}{3}\right)}^{\pi}
 \end{aligned}$$

$$\text{We know } 2+3\cos\theta=0 \Rightarrow \cos\theta=-\frac{2}{3}$$

$$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{4}{9}} = \sqrt{\frac{5}{9}} = \frac{\sqrt{5}}{3}$$

$$\text{ii Area of smaller loop} = \left\{ \frac{17(\pi)}{2} - 0 + 0 \right\} - \left\{ \frac{17 \sin^{-1} \frac{2}{3}}{2} - \frac{9(\frac{2}{3})(\frac{\sqrt{5}}{3}) + 12 \frac{\sqrt{5}}{3}}{2} \right\}$$

$$= \frac{17\pi}{4} - \frac{17}{2} \sin^{-1} \frac{2}{3} + \sqrt{5} - 4\sqrt{5}$$

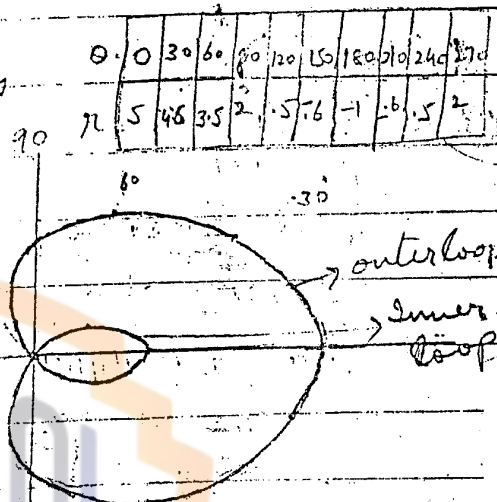
$$= \frac{17\pi}{4} - \frac{17}{2} \sin^{-1} \frac{2}{3} - 3\sqrt{5}$$

$$\textcircled{23} r = 2 + 3 \cos \theta$$

$$\text{Put } r = 0$$

$$-\frac{2}{3} = \cos \theta$$

$$\theta = \cos^{-1}(-\frac{2}{3}) = 131.48^\circ$$



The upper half of the outer loop is determined by

$$\theta = 0 \text{ to } \theta = \cos^{-1}(-\frac{2}{3})$$

and lower half of the smaller loop is traced by

$$\theta = \cos^{-1}(-\frac{2}{3}) \text{ to } \theta = \pi$$

Area between the outer loop & inner, i.e. shaded area

$$= \text{Area of outer loop} - \text{Area of inner smaller loop}$$

$$= 2 \int_0^{\cos^{-1}(-\frac{2}{3})} \frac{r^2}{2} d\theta - 2 \int_{\cos^{-1}(-\frac{2}{3})}^{\pi} \frac{r^2}{2} d\theta$$

7.5-29

∴ Area between the outer loop + inner loop =

$$\left[\frac{17 \cos^{-1}(-\frac{2}{3})}{2} + \frac{9}{2} \cdot \frac{\sqrt{5}}{3} \left(-\frac{2}{3}\right) + 12 \frac{\sqrt{5}}{3} - 0 \right] - \left[\frac{17\pi}{2} + 0 + 0 - \frac{17 \cos^{-1}(-\frac{2}{3})}{2} \right]$$

$$- \frac{17 \cos^{-1}(-\frac{2}{3})}{2} - \sqrt{5} + 4\sqrt{5} - \frac{17\pi}{2} + \frac{17 \cos^{-1}(-\frac{2}{3})}{2} - \sqrt{5} + 4\sqrt{5}$$

$$= 17 \cos^{-1}(-\frac{2}{3}) + 3\sqrt{5} - \frac{17\pi}{2} + 3\sqrt{5}$$

$$= 17 \cos^{-1}(-\frac{2}{3}) - \frac{17\pi}{2} + 6\sqrt{5} \quad \text{Ans.}$$

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